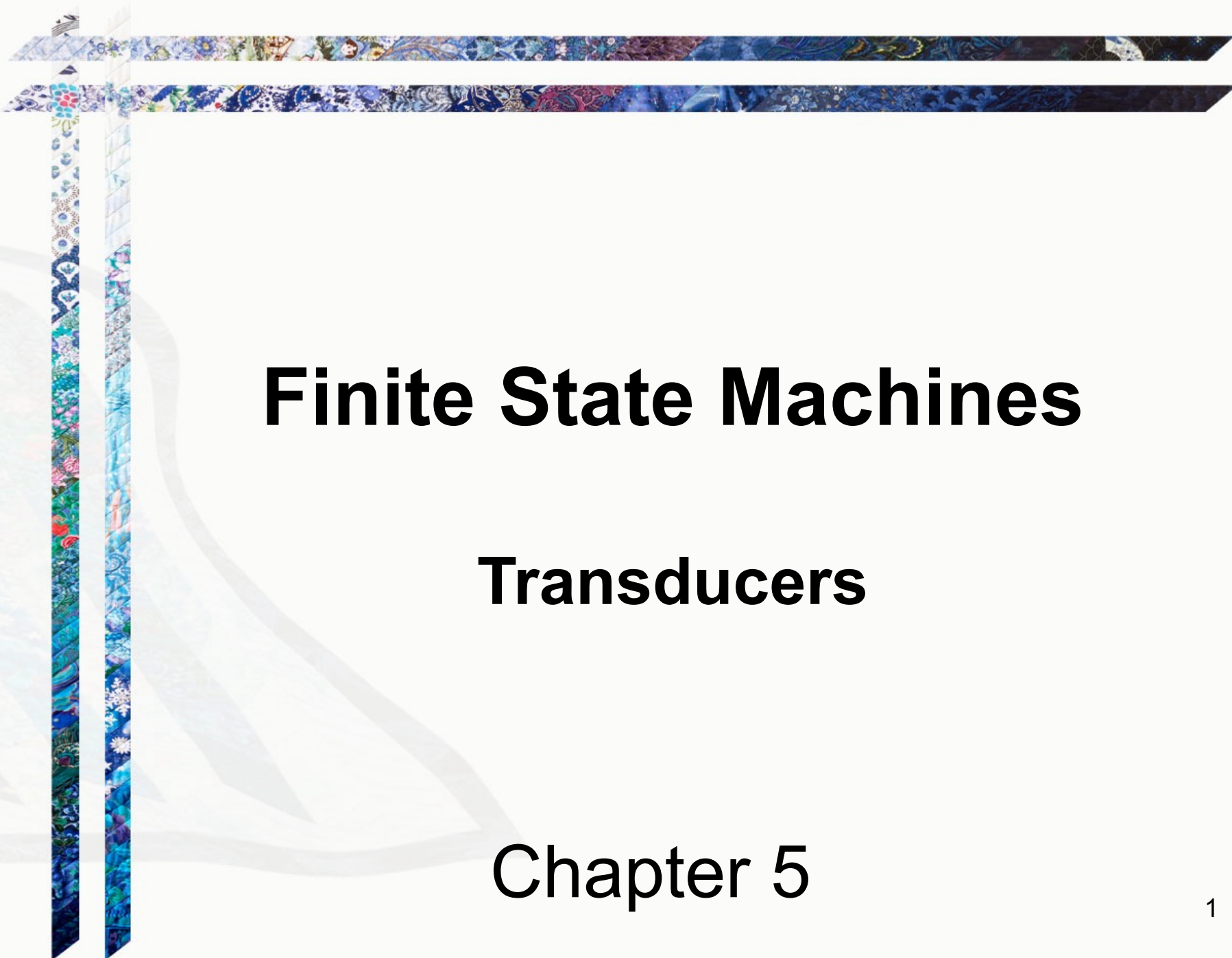




Finite State Machines

Transducers

Chapter 5

Deterministic Finite State Transducers

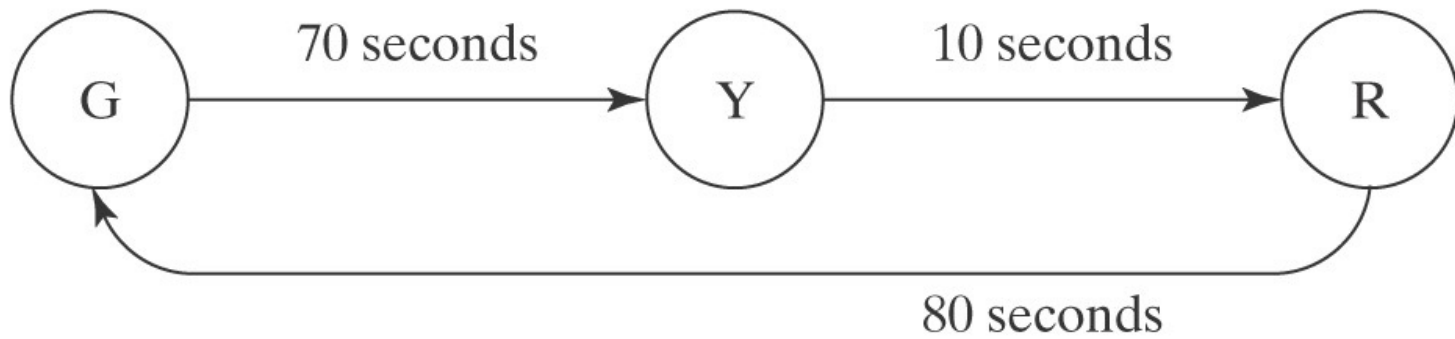
A **Moore machine** $M = (K, \Sigma, O, \delta, D, s, A)$, where:

- K is a finite set of states
- Σ is an input alphabet
- O is an output alphabet
- $s \in K$ is the initial state
- $A \subseteq K$ is the set of accepting states,
- δ is the transition function from $K \times \Sigma$ to K ,
- D is the **output function** from K to O^* .

M outputs each time it lands in a state.

A Moore machine M computes a function $f(w)$ iff, when it reads the input string w , its output sequence is $f(w)$.

A Simple US Traffic Light Controller



Deterministic Finite State Transducers

A **Mealy machine** $M = (K, \Sigma, O, \delta, s, A)$, where:

- K is a finite set of states
- Σ is an input alphabet
- O is an output alphabet
- $s \in K$ is the initial state
- $A \subseteq K$ is the set of accepting states
- δ is the transition function from $K \times \Sigma$ to $K \times O^*$

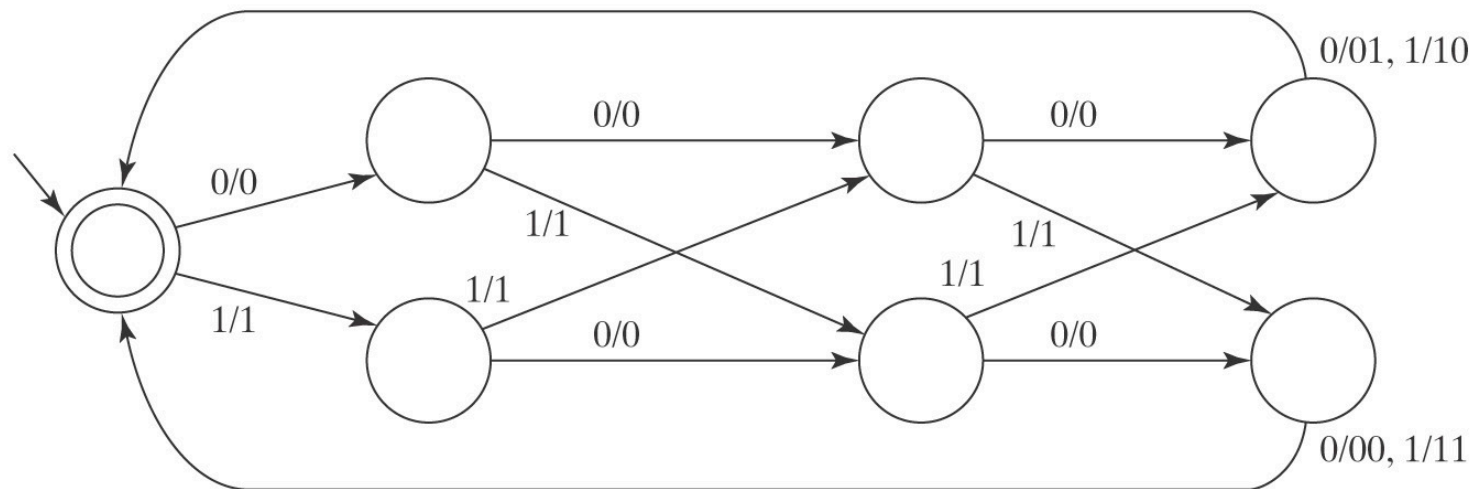
M outputs each time it takes a transition.

A Mealy machine M computes a function $f(w)$ iff, when it reads the input string w , its output sequence is $f(w)$.

An Odd Parity Generator

After every four bits, output a fifth bit such that each group of four bits has odd parity.

0 0 1 0 1 1 0 0 0 0 0 0 1 1 1 1

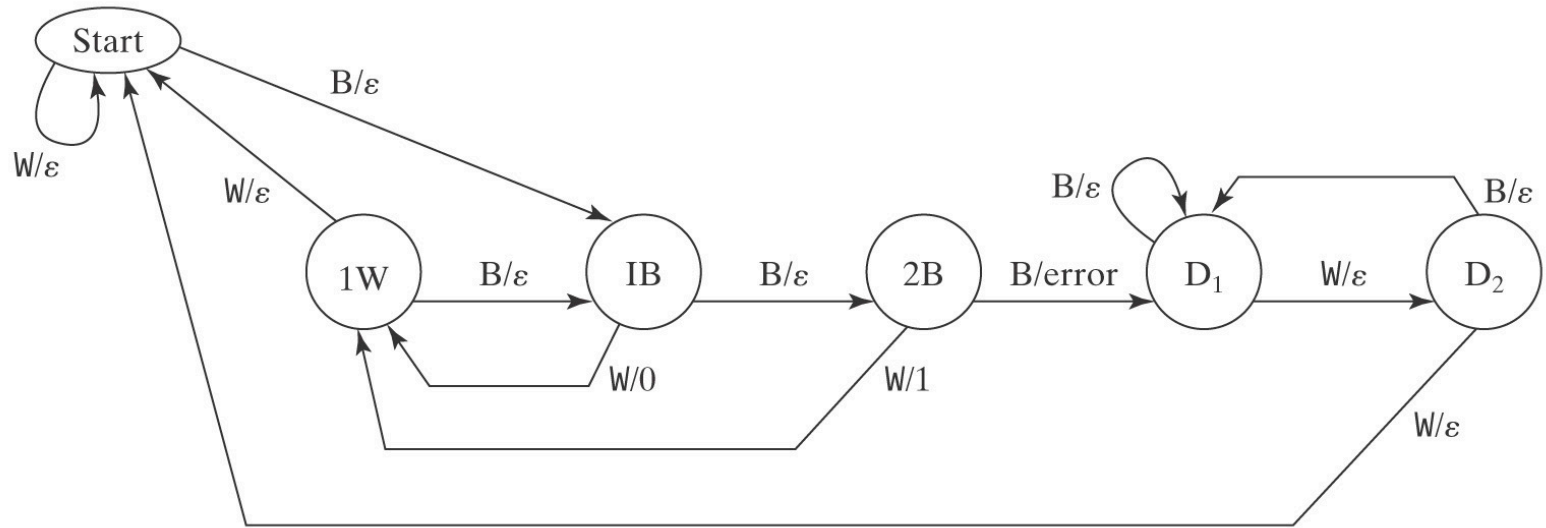


input / output

A Bar Code Scanner



A Bar Code Scanner



English Morphology

